Design and Analysis of a Multi-port DC-DC Converter for Green Hydrogen Production

ABSTRACT

The growing demand for clean and sustainable energy has increased the importance of green hydrogen production systems. This project presents the design and analysis of a multi-port DC-DC converter using MATLAB/Simulink for green hydrogen applications. The converter integrates multiple power sources including a solar DC source and battery backup to provide stable DC output to an electrolyzer load. The designed system demonstrates effective voltage regulation, energy transfer, and stable converter operation. Output waveforms and simulation results validate the performance of the proposed converter system.

INTRODUCTION

Green hydrogen is considered one of the most promising clean energy sources for future energy systems. It is produced through electrolysis using renewable energy resources such as solar and wind power. DC-DC converters play an important role in regulating and controlling power flow between renewable sources and hydrogen electrolyzers.

A multi-port DC-DC converter combines multiple energy sources into a single power conversion system. In this project, a solar DC source and battery are integrated using a converter topology to supply stable power to an electrolyzer load. MATLAB/Simulink is used to model and analyze the converter performance.

OBJECTIVES

The main objectives of this project are:

·To design a multi-port DC-DC converter using MATLAB/Simulink.
·To integrate solar and battery energy sources.
·To analyze converter output characteristics.
·To study voltage regulation and system stability.
·To simulate converter response for green hydrogen production applications.

Theory of Multi-port DC-DC Converter

A DC-DC converter is an electronic power conversion device used to convert one level of DC voltage into another level of DC voltage. In renewable energy systems, DC-DC converters are essential for maintaining stable voltage and improving energy transfer efficiency.

A multi-port DC-DC converter consists of multiple input power sources connected to a common converter structure. The main advantage of a multi-port converter is its ability to manage power from different sources simultaneously. In this project, the converter combines energy from:

- Solar DC source
- Battery backup source

The solar source acts as the primary renewable energy supply, while the battery provides backup energy during fluctuations or insufficient solar power conditions.

The converter uses semiconductor switching devices such as MOSFETs along with passive components like inductors and capacitors to regulate output voltage. The switching operation is controlled using Pulse Width Modulation (PWM) signals generated by the pulse generator block.

The inductor stores energy during switching intervals and releases it to the load, while the capacitor smooths the output voltage and reduces ripple. The diode ensures unidirectional current flow and protects the circuit during switching operation.

The regulated output voltage can then be supplied to an electrolyzer system for green hydrogen production.

Working Principle

The working operation of the proposed multi-port DC-DC converter can be divided into different stages.

Initially, the solar DC source and battery source provide input power to the converter circuit. The MOSFET acts as a high-speed electronic switch controlled by the pulse generator using PWM signals.

When the MOSFET is switched ON:

- Current flows through the inductor.
- Energy is stored in the magnetic field of the inductor.
- The capacitor begins charging.

When the MOSFET is switched OFF:

- The stored energy in the inductor is released to the load through the diode.
- The capacitor helps maintain stable output voltage.
- The resistor acts as the electrolyzer load representation.

The continuous switching process regulates the output voltage and ensures stable power transfer. The voltage sensor measures output voltage, and the scope displays waveform characteristics for analysis.

The obtained waveform shows transient response during initial switching operation followed by stable steady-state output, demonstrating proper converter functionality.

Theory of Green Hydrogen Production

Green hydrogen is produced through the electrolysis process, where water molecules are split into hydrogen and oxygen using electrical energy obtained from renewable energy sources.

The electrolysis reaction can be represented as:

$$2H2O \rightarrow 2H2 + O2$$

Unlike conventional hydrogen production methods that use fossil fuels, green hydrogen generation produces minimal environmental pollution and supports sustainable energy development.

For efficient electrolysis, a stable DC power supply is required. Therefore, multi-port DC-DC converters are important components in modern hydrogen energy systems because they help integrate renewable energy sources and maintain regulated output voltage for electrolyzer operation.

Output Analysis

The simulation output waveform obtained from the scope demonstrates the dynamic behavior of the converter. Initially, the converter experiences transient oscillations and voltage overshoot due to switching action and energy transfer between inductive and capacitive components.

After a short transient period, the output voltage gradually stabilizes and reaches steady-state operation. This indicates that the converter is capable of maintaining regulated DC output suitable for hydrogen production applications.

The MATLAB analysis also validates the theoretical response of the converter system and confirms stable converter performance under specified operating conditions.

COMPONENTS USED

- **DC Voltage Source** – Used to represent the solar photovoltaic energy source supplying renewable DC power to the converter.
- **Battery** – Acts as a backup energy source and supports continuous power supply during low solar conditions.
- **MOSFET** – Functions as the main high-speed switching device in the DC-DC converter circuit.
- **Pulse Generator** – Generates PWM switching pulses to control MOSFET operation.
- **Diode** – Allows current flow in one direction and prevents reverse current flow.
- **Inductor** – Stores energy in magnetic form and helps in voltage conversion.
- **Capacitor** – Filters output voltage and reduces ripple to maintain stable DC output.
- **Resistor** – Represents the electrolyzer load used for green hydrogen production.
- **Voltage Sensor** – Measures output voltage of the converter system.
- **PS-Simulink Converter** – Converts physical signals into Simulink signals for waveform analysis.
- **Scope** – Displays output voltage waveform and converter response.
- **Electrical Reference (Ground)** – Provides common reference point for the electrical circuit.
- **Solver Configuration Block** – Ensures proper simulation and operation of the Simscape electrical network.

CIRCUIT DESIGN

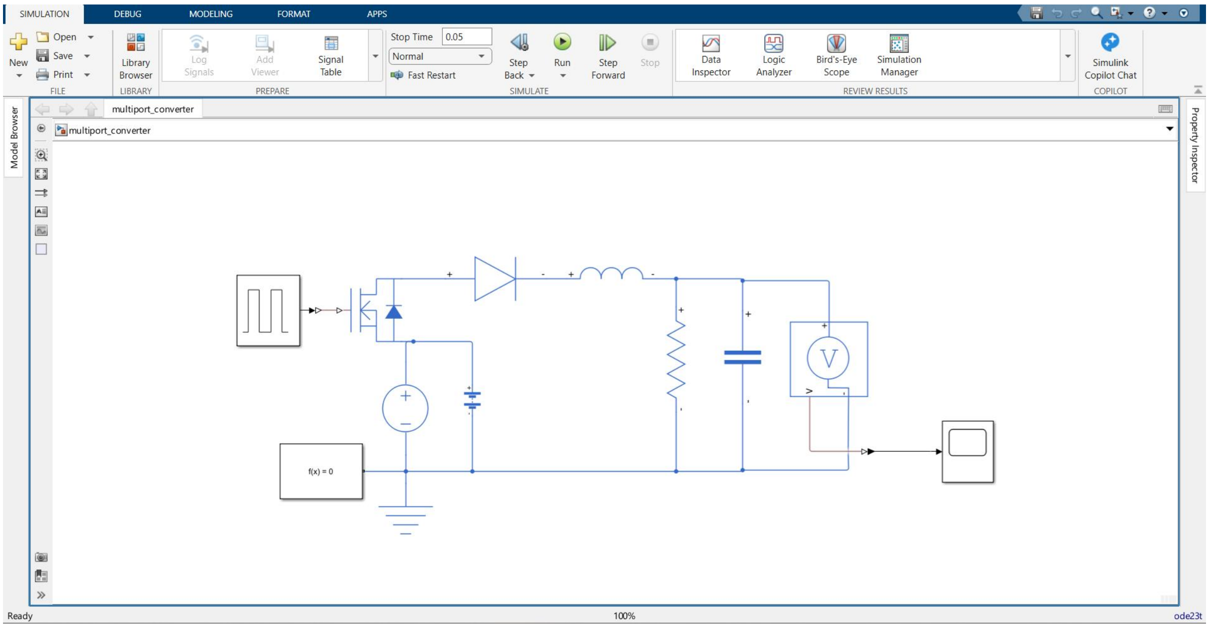

Figure 1: Simulink model of the proposed multi-port DC-DC converter system.

MATLAB ANALYSIS

MATLAB scripting was additionally used to analyze converter output characteristics and validate theoretical output response. Converter parameters such as source voltage, switching frequency, inductance, capacitance, and load resistance were defined within the MATLAB environment.

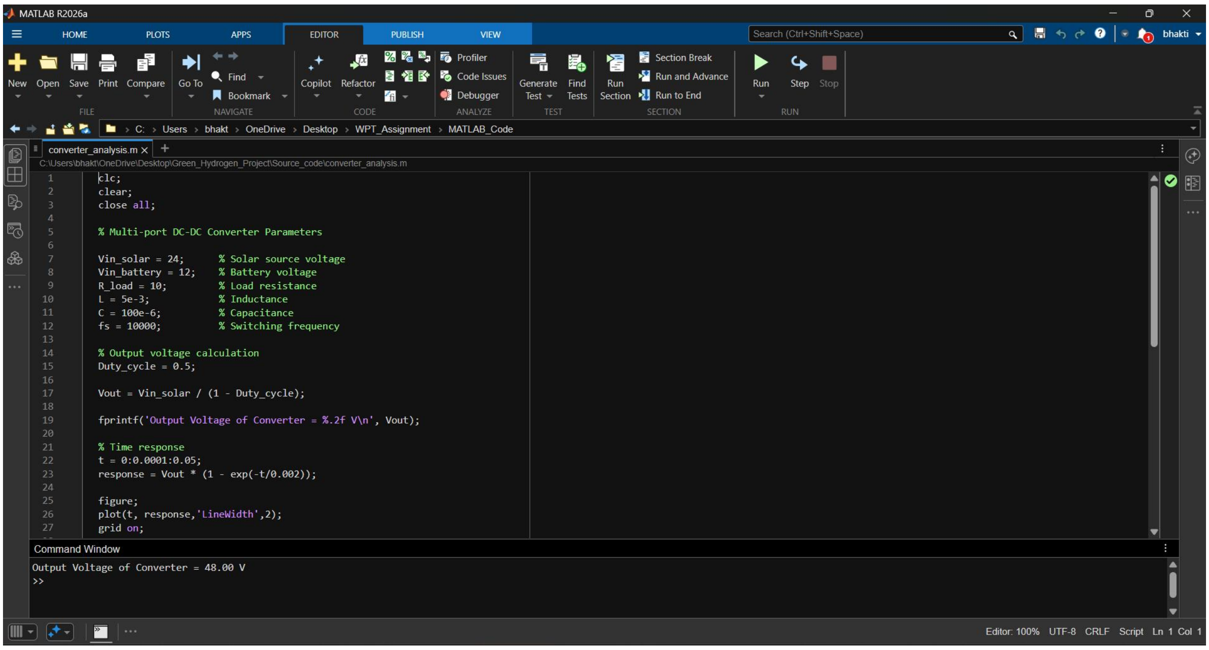

Figure 2: MATLAB analysis code for converter response.

OUTPUT RESULTS

The simulation output waveform shows the transient response and stabilization behavior of the converter. Initially, the converter experiences a small overshoot and oscillation due to switching dynamics. After the transient region, the output voltage becomes stable, indicating proper converter operation and voltage regulation.

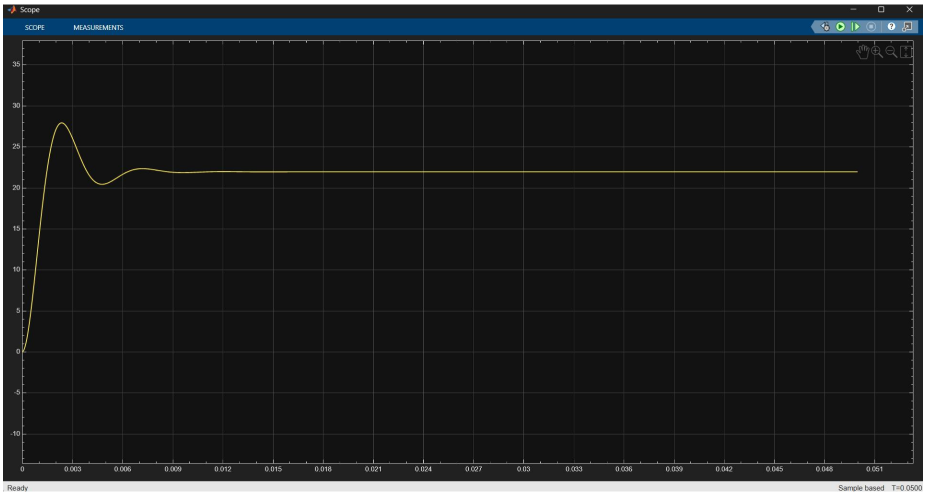

Figure 3: Output voltage waveform obtained from Simulink scope.

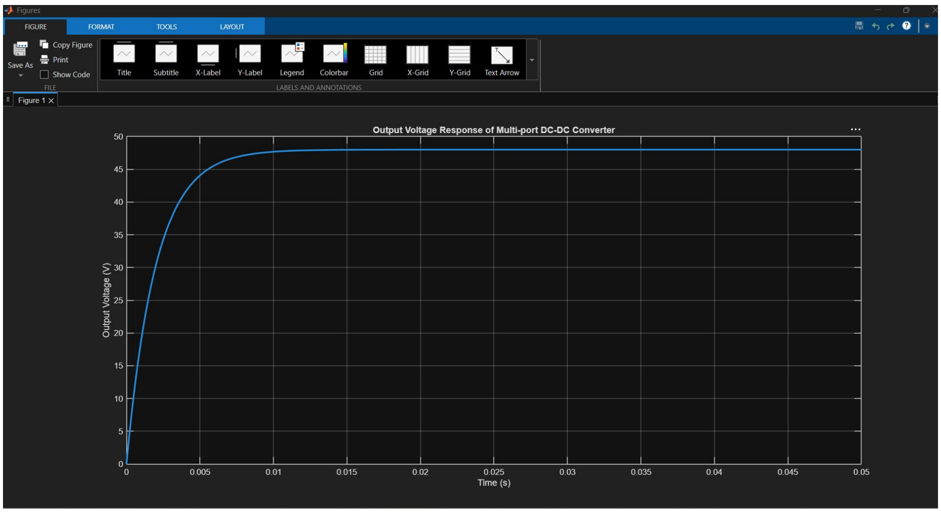

Figure 4: MATLAB-generated output response graph.

ADVANTAGES

- Efficient utilization of renewable energy.
- Stable DC output voltage.
- Integration of multiple power sources.
- Improved energy management.
- Suitable for green hydrogen systems.

APPLICATIONS

- Green hydrogen production systems
- Renewable energy integration
- Smart energy management
- Electric vehicle charging systems
- Industrial DC power systems

CONCLUSION

The project successfully demonstrates the design and analysis of a multi-port DC-DC converter for green hydrogen production applications using MATLAB/Simulink. The converter effectively integrates solar and battery sources while maintaining stable output voltage. Simulation results validate the converter performance and demonstrate its suitability for renewable energy applications.

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